Monohybrid Cross Worksheet

Answer each of the following questions using a Punnett Square and the rules of monohybrid crosses.

1.) The allele for dimples (D) is dominant to the allele for no dimples (d). A man heterozygous for dimples marries a woman who is also heterozygous for dimples.

Determine all potential dimple genotypes and phenotypes for the offspring of this man and woman.

Parents
Phenotype — Dimples — Dimples
Genotype — Dd — Dd
Gametes — D d — D d
F1 generation

	D	d
D	DD	Dd
d	Dd	dd

75% dimples (DD, Dd): 25% no dimples (dd)

2.) The allele for hitchhiker's thumb (h) is recessive to straight thumb (H). If a man and his wife are both homozygous recessive, will any of their offspring potentially have hitchhikers thumb?

Determine all potential hitchhiker's thumb genotypes and phenotypes for the offspring of this man and woman. Is it possible for any offspring of the F_1 generation to have hitchhiker's thumb?

Parents
Phenotype — hitchhikers thumb — hitchhikers thumb
Genotype — hh — hh
Gametes — h — h
F1 generation

	h	h
h	hh	hh
h	hh	hh

100% hh hitchhikers thumb

3.) In a certain breed of dogs, a gene (L) codes for hair length. The dominant trait is short hair and the recessive is long hair. Suppose a heterozygous female dog and a homozygous recessive male dog mate.

Determine all the potential offspring of the F_1 generation.

Parents
Phenotype long short
Genotype ll Ll
Gametes l L l
F1 generation

	l	l
L	Ll	Ll
l	ll	ll

50% Ll short: 50% ll long

4.) In fruit flies, the allele for normal wings (V) is dominant to the allele for short wings (v). Suppose two fruit flies heterozygous for the trait are mated.

Determine all the potential offspring of the F_1 generation.

Parents
Phenotype normal normal
Genotype Vv Vv
Gametes V v V v
F1 generation

	V	v
V	VV	Vv
v	Vv	vv

75% normal (VV, Vv) : 25% short (vv)

5.) A genetic engineer is going to cross two watermelon plants to produce seeds for a spring planting. He is breeding for size, and wants to have as many watermelons with the phenotype for long shape as possible. In watermelons, the allele for short shape (R) is dominant to the allele for long shape (r). Would crossing a watermelon homozygous recessive for the trait with a watermelon

heterozygous for the trait give the most long watermelons possible? Explain your answer using Punnett Squares.

Parents — Phenotype: short — long
Genotype: Rr — rr
Gametes: R r — r r
F1 generation

	R	r
r	Rr	rr
r	Rr	rr

50% short Rr: 50% long rr

Parents — Phenotype: short — short
Genotype: Rr — Rr
Gametes: R r — R r
F1 generation

	R	r
R	RR	Rr
r	Rr	rr

75% short Rr, RR: 25% long rr

Parents — Phenotype: long — long
Genotype: rr — rr
Gametes: r r — r r
F1 generation

	r	r
r	rr	rr
r	rr	rr

100% long rr